MILITARY SPECIFICATION

WINCH, DRUM, POWER OPERATED WITH SEPARATE GYPSY DRIVE (SPANWIRE WINCH, TYPE 15.5-SW-1D/6.0 - 2G)

1. SCOPE

1.1 Scope. This specification covers the detail requirements for type 15.5-SW-1D/6.0-2G single drum spanwire winch rated at 15,500 pounds at 240 feet per minute with a separate gypsy drive rated 6,000 pounds at 60/120 feet per minute.

2. APPLICABLE DOCUMENTS

2.1 The following specifications of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard.
MIL-W-15802 - Winch, Gypsy, Power Operated, Electric.
MIL-B-16392 - Brakes, Magnet, Naval Shipboard.
MIL-M-17060 - Motors, 60 cycle, Alternating-Current, Integral Horsepower, (Shipboard Use).
MIL-W-17265 - Winch, Drum, Power Operated.
MIL-C-23742 - Control Centers, Alternating Current, Naval Shipboard.

(Copies of specifications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. Winch, drum, power operated, type 15.5-SW-1D/6.0-2G shall be in accordance with MIL-W-17265, except as specified herein.

3.2 Construction. Winch Type 15.5-SW-1D/6.0-2G shall consist of two independent winches mounted integrally on a single bedplate. Each unit shall be independently operated and shall have its own power unit and operating controls.

3.2.1 Drum unit. The drum unit shall be in accordance with MIL-W-17265, except for gypsy heads and light line speed, hoisting and lowering, of not less that 180 percent rated full load hoisting speed, are not applicable.

3.3 Drums. The drum for winch type 15.5-SW-1D/6.0-2G shall be designed for the following:

(a) Stowage capacity for 800 feet of 7/8-inch wire rope. The breaking strength of the wire rope is 65,800 pounds.

(b) Barrel diameter of 30-inches, plus or minus 1/2-inch.

(c) Distance between flanges shall be 15-inches, plus or minus 1/2-inch.

3.3.1 The winch drum shall be supported on anti-friction bearings on the gypsy head shaft; and the drum shall be located as near the midlength of the shaft as practical. Design shall allow independent operation of the drum and gypsy units.

3.4 Gypsy winch. The gypsy unit of winch type 15.5-SW-1D/6.0-2G shall be in accordance with MIL-W-15802 unless otherwise specified in this specification.

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3.4.1 Gypsy winch electric motor. The gypsy winch electric motor shall be in accordance with MIL-W-17060 and the following:

Classification	Navy A
Ambient temperature	40 degrees C.
Voltage	440 AC
Phase	3
Cycles per second	60
Duty	Full load, 1/2 hour; 1/2 load, continuous
Enclosure	Watertight
Speed	Two speed (full and half) constant torque
Type	Squirrel cage induction
Design	D
Horsepower	As required

3.4.2 Gypsy winch motor controller. The gypsy winch electric motor controller and master switch shall be in accordance with MIL-C-2212 or MIL-C-23742 as applicable and the following:

Operation	Magnetic
Voltage	440 VAC 3 phase 60 cycles
Type	Across the line
Performance	Non-automatic
Function	Motor starting, reversing and speed selection
Protection	Low voltage release and thermal overload
Enclosure	Dripproof protected
Duty	Continuous
Emergency run feature	Remote emergency run pushbutton shall be provided with a watertight housing
Ambient temperature	50 degrees C.
Master switch	Remote drum type, lever operated spring return to OFF, watertight enclosure with power ON indicating light

3.4.3 Gypsy winch electric brake. The gypsy winch electric brake shall be in accordance with MIL-B-16392 and the following:

Voltage	440 AC
Duty	Continuous
Ambient temperature	40 degrees C.
Enclosure	Watertight
Mounting	As required

3.5 Gearing. The gear drives shall have a service factor of 1.25.

3.6 Controlled spooling.

3.6.1 The drum shall be provided with controlled spooling. The controlled spooling shall insure that the wire spools on the drum in an evenly spaced, interrupted spiral pattern from flange to flange. The first and each succeeding layer of wire on the drum shall form a spooling pattern for the next layer that will exactly reproduce this spooling pattern on all layers. The spooling pattern of wire on the drum shall eliminate overriding (cross-over) turns and under-cutting of the wire rope and result in a minimum amount of scrubbing and chaffing of the wire rope.

3.6.2 A power operated level wind shall be provided to insure that the wire maintains the proper position with respect to the drum at all times.

3.7 Anti-slack device. Each drum provided with controlled spooling shall also be provided with an anti-slack device. The anti-slack device shall prevent slack wire from developing between the winch drum and the anti-slack device when it is turned on. A direct lead from the spooling device to the ram tensioner via the anti-slack device shall be provided. No intermediate sheaves shall be used.

3.7.1 The anti-slack device shall be hydraulically driven and shall be connected to the swivel sheave of the swiveling fairleader.

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3.7.2 The anti-slack devices for type 15.5-SW-1D/6.0-2G winches shall develop a line pull of 750 pounds at all winch power payout speeds from zero to maximum. The anti-slack devices shall also develop a drag of 1500 pounds when wire is hauled in at all speeds from zero to maximum.

3.7.3 Hydraulically actuated pressure rollers shall be provided to insure positive contact between the wire and the drive sheave on the swiveling fairleader sheave.

3.7.4 A snap action "On-Off" switch shall be mounted as part of the control console to permit the operator to activate or deactivate the anti-slack device.

3.7.5 When the anti-slack device is deactivated the pumping unit and the hydraulic motor shall be disengaged or be otherwise protected and the pressure rollers retracted.

3.7.6 When the anti-slack device is activated the load on the winch will be light, and therefore the power required to operate it shall not be considered as additive to winch rated load power requirements for determination of winch component size such as electric motor.

3.8 Rope lead. The wire rope lead on winch drums provided with controlled spooling and anti-slack devices shall be from the level wind to the swiveling fairleader.

3.9 Rated line pull and line speed. Rated line pull and line speed for winch type 15.5-SW-1D/6.0-2G shall be as follows:

(a) Drum Unit
  (1) Rated line pull - 15,000 pounds on the average layer of the drum;
  (2) Rated line speed - 240 feet per minute

(b) Gypsy Unit
  (1) Rated line pull - 6000 pounds
  (2) Rated line speed - 120 and 60 feet per minute

3.10 Duty. The drum unit of winch type 15.5-SW-1D/6.0-2G shall be designed to permit continuous operation for a period of six hours. A normal operating cycle will consist of 10 seconds at rated speed, rated load; 20 seconds at idle; 10 seconds at half speed and rated load; and 20 seconds at idle shall be used during this six hour period. The duty for the gypsy unit shall be as specified in paragraph 3.4.1.

3.10.1 Name plate ampere rating of the electric motors for both the drum and gypsy units shall not be exceeded by more than twenty-five percent when the winch is being operated at rated load and rated speed (constant velocity), temperature rise shall not exceed nameplate permissible temperature rise when the winch is operated at the duty cycle and period specified in 3.10.

3.10.2 Thermal overload, for the drum unit, shall be indicated by energizing a master panel indicator light through a thermal heating device (see thermal requirements of MIL-W-17265) in either the motor or the motor controler. The indicator light shall be in accordance with the illumination requirements of MIL-W-17265. The device shall not de-energize the spanwire winch unit motor.

3.11 Size and Weight. The winch shall be as compact and light as is consistent with adequate strength and rigidity. Weight and size shall not exceed the following:

(a) Maximum wieght - 19,000 pounds
(b) Maximum width (across drum) with gypsy heads installed - 102 inches
(c) Maximum length of winch, excluding anti-slack devices - 140 inches
(d) Maximum height - 74 inches

4. QUALITY ASSURANCE PROVISIONS

4.1 In addition to the tests specified in 17265 the following torque clutch tests shall be conducted.

4.2 Torque clutch test. Torque clutch tests shall be made at the manufacturer's plant.

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4.2.1 Each drum torque clutch shall be tested by pulling a 20-foot length of wire off the drum at a speed of 240 feet per minute. The maximum tension measured during this test shall be recorded. (The use of a special locking device on the winch to hold the drum shaft stationary during this test is acceptable.)

4.2.2 The torque clutch of one winch of each type in the lot shall be tested by repeating the above test two times, waiting approximately 15 seconds between runs, and seven additional times, allowing the clutch to cool as necessary between runs. The maximum tension measured during each run shall be recorded.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-W-17265.

6. NOTES

6.1 Ordering data. Ordering data shall be as specified in section 6 of MIL-W-17265.

Preparing activity:
Navy - SH
(Project 3950-N0120)

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